Computational aspects of penalized least-squares trend estimation with controlled smoothness

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Abstract

Penalized least-squares trend estimators provide an interpretable way to separate persistent variation from short-term fluctuations. Their practical use nevertheless requires choices about difference order, reference differences, smoothing intensity, and numerical implementation. We examine these choices through a reproducible computational framework for finite-difference penalties. We derive the polynomial null space and the spectral representation of the smoother, and normalize the trace-based smoothness index so that its range is independent of the difference order. We also study the conditional profile estimator of a nonzero reference-difference mean, distinguishing this derivation from procedures available in the software. Controlled experiments compare naive and cached dense linear systems, a reusable singular-value decomposition, and a banded solver. We separately investigate repeated full-trend reconstruction and scalar effective-degrees-of-freedom evaluations, including numerical agreement and preprocessing time. A nonlinear synthetic example illustrates the behavior of classical smoothing criteria, while contemporary NOAA atmospheric CO_2 and monthly US federal funds observations support auditable, multi-panel comparisons across four differencing orders, CV, GCV, AICc, BIC, Gaussian profile MLE, prescribed smoothness, and fixed or data-derived reference means. We distinguish historical single-origin holdout extrapolations from unvalidated future trend extensions. The results highlight workload-dependent computational trade-offs rather than the universal superiority of any individual algorithm. The accompanying Python implementation and executable experiments provide a transparent basis for further numerical and statistical investigation.

Keywords: Whittaker–Henderson smoothing, spectral decomposition, smoothing parameter selection, time-series trend, reproducible computation

1 Introduction

An observed time series may combine low-frequency behavior with temporary movements and measurement noise. Estimating its trend is a descriptive task: the resulting curve depends on how fidelity to the observations is traded against smoothness. Whittaker's graduation framework and subsequent finite-difference penalized least-squares (PLS) estimators formalize this compromise Whittaker (1923); Guerrero (2007). The Hodrick–Prescott filter is an important second-difference special case, but the statistical problem admits other penalty orders and nonzero reference values for the differences Cortés-Toto et al. (2017).

The tuning parameter must also be chosen. Cortés-Toto, Guerrero, and Reyes Cortés-Toto et al. (2017) investigate attained smoothness when choosing the penalty with cross-validation (CV), generalized cross-validation (GCV), corrected Akaike information criterion (AICc), or Bayesian information criterion (BIC). Their simulations vary trend shape, series length, variability, and seasonality. Efficient Whittaker–Henderson computation has an extensive history, including methods exploiting banded factorization and structure to reduce numerical expense Weinert (2007); consequently, diagonalization is not presented here as a newly discovered mathematical algorithm.

In the scientific-computing literature, Balam et al. Balam et al. (2025) compare normal equations, QR, and singular-value decomposition (SVD) in least-squares computation and discuss conditioning. Azofeifa and Moreles Azofeifa and Moreles (2025) present a different numerical model, but similarly distinguish the formulation, solver, numerical comparisons, and limitations. We similarly distinguish statistical formulation, numerical algorithms, empirical comparisons, and their limitations.

Our objective is to provide a small, auditable Python package, `trendestimation`, that implements the *published* PLS family and exposes examples as executable notebooks within Sphinx documentation. Specifically, we describe the affine solution for general order d and reference mean μ, its spectral implementation, the criteria available in the public API, and numerical experiments that report computational time, solver agreement, and precomputation costs alongside reproducible illustrations. Experimental forecast-horizon cross-validation and dynamic tracking of minima from other research projects are not included. The distinction between an in-sample smoother and an ex ante forecast is maintained throughout.

2 Model and spectral representation

2.1 Finite-difference penalized least squares

Let $y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$ denote an equally spaced series. For an integer $1 \le d < n$, let $D_d \in \mathbb{R}^{(n-d)\times n}$ represent the dth forward-difference operator; for example, row entries of D_2 are $(1, -2, 1)$. We consider the published estimator

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau\in\mathbb{R}^n} \left\{ \|y - \tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}, \qquad \lambda \ge 0, \tag{1}$$

where $\mu \in \mathbb{R}$ is a fixed reference mean for the differences. The choice $\mu = 0$ gives the familiar pure penalty. The implementation additionally permits $d = 0$, with $D_0 = I_n$, as a well-defined edge case; the geometric statements below concern $1 \le d < n$.

Write $Q_d = D_d^\top D_d$ and choose $p \in \mathbb{R}^n$ satisfying $D_d p = \mu \mathbf{1}$. Using zero-based indices $t = 0, \dots, n-1$, one convenient choice is $p_t = \mu \binom{t}{d}$ because $D_d \binom{t}{d} = 1$. Setting $x = \tau - p$ and differentiating the strictly convex objective gives

$$\widehat{\tau}_{\lambda,\mu} = p + H_\lambda (y - p), \qquad H_\lambda = (I_n + \lambda Q_d)^{-1}. \tag{2}$$

Indeed, $(I_n + \lambda Q_d)x = y - p$ and $I_n + \lambda Q_d$ is positive definite for every finite $\lambda \ge 0$. This affine expression is equivalent to the normal equations $(I_n + \lambda Q_d)\widehat{\tau} = y + \lambda \mu D_d^\top \mathbf{1}$.

2.2 Proof of the *d*-dimensional kernel

For $1 \le d < n$, the dth forward difference has the explicit binomial stencil

$$(D_d \tau)_t = \sum_{k=0}^{d} (-1)^{d-k} \binom{d}{k} \tau_{t+k}, \quad t = 0, \dots, n-d-1. \tag{3}$$

The first $n-d$ columns of D_d form an upper triangular $(n-d) \times (n-d)$ submatrix with nonzero diagonal $(-1)^d$. Hence $\operatorname{rank}(D_d) = n - d$. As $Q_d = D_d^\top D_d$, $x^\top Q_d x = \|D_d x\|^2$, and therefore

$$\ker Q_d = \ker D_d, \qquad \dim \ker Q_d = n - \operatorname{rank}(D_d) = d. \tag{4}$$

The vectors $1, t, \dots, t^{d-1}$ have vanishing dth forward differences and are linearly independent on $n > d$ nodes. Thus they give a basis of the kernel, proving that exactly d eigenvalues of Q_d vanish in exact arithmetic. Under $\mu = 0$, the limit $\lambda \to \infty$ is the least-squares projection onto polynomials of degree $\le d - 1$. For instance $d = 1$ preserves a constant, $d = 2$ a line, and $d = 3$ a quadratic. When $\mu \ne 0$, the limiting set is instead the affine space $\{\tau : D_d \tau = \mu \mathbf{1}\}$, which has the same d homogeneous directions but includes a degree-d particular solution. In the implementation $p_t = \mu \binom{t}{d}$.

2.3 Interpretation of higher orders and the spline connection

For a sufficiently smooth function f observed on an equally spaced mesh of width h, Taylor expansion yields $\Delta_h^d f(x_t) = h^d f^{(d)}(x_t) + O(h^{d+1})$. Consequently, under appropriate mesh-refinement assumptions,

$$h^{1-2d} \sum_t (\Delta_h^d f(x_t))^2 \longrightarrow \int |f^{(d)}(x)|^2 \, dx. \tag{5}$$

This is a mathematical bridge to roughness penalties in smoothing splines. A Whittaker-type trend smoother penalizes differences of fitted ordinates directly, while a P-spline instead penalizes differences of coefficients in a chosen spline basis; these

are related ideas but not identical estimators. For a continuous penalty weight λ_{cont}, the corresponding discrete coefficient is scaled approximately as $\lambda_{\text{disc}} = \lambda_{\text{cont}} h^{1-2d}$.

The order d is an integer representing the order of the derivative being approximated; it is not a continuous interpolation between derivative orders. Comparing different d values is a change of regularization model, so equal numerical λ does not imply equal smoothing. It is useful to examine orders $d = 1, \ldots, 5$ under a common normalized trace-based index; however, arbitrary orders and polynomial nullspaces have long been known Weinert (2007); Biessy (2026). Larger d can be ill-conditioned because the binomial stencil has growing coefficients and the positive eigenvalues may span many orders of magnitude. Neither higher d nor higher polynomial degree guarantees a better trend or numerical stability.

2.4 Null space, SVD, and limiting trends

The rows of D_d are independent in exact arithmetic, so $\operatorname{rank}(D_d) = n - d$ and $\dim \ker D_d = d$. The null space consists of the sampled polynomials of degree at most $d-1$. Let $D_d = U \Sigma V^\top$ be a full SVD. With nonnegative eigenvalues δ_j of Q_d, ordered so that exactly d are zero, the smoother becomes

$$H_\lambda = V \operatorname{diag}\left(\frac{1}{1+\lambda\delta_1}, \ldots, \frac{1}{1+\lambda\delta_n} \right) V^\top. \tag{6}$$

Therefore the transformed observations $c = V^\top (y - p)$ can be computed once, and every new penalty value changes only the scalar attenuation factors $(1 + \lambda\delta_j)^{-1}$. At $\lambda = \infty$ the exact limiting fit is $p + P_{\ker D_d}(y - p)$, rather than a computation using an arbitrarily large finite penalty.

For fixed μ, the effective degrees of freedom are $\operatorname{EDF}(\lambda) = \operatorname{tr}(H_\lambda) = \sum_j (1 + \lambda\delta_j)^{-1}$. We report Guerrero's trace-based smoothness $S_G = 1 - \operatorname{tr}(H_\lambda)/n$ alongside the normalized coordinate

$$S(\lambda) = \frac{n - \operatorname{tr}(H_\lambda)}{n - d} \in [0, 1], \quad S(0) = 0, \quad S(\infty) = 1. \tag{7}$$

For $1 \le d < n$, this coordinate is strictly increasing for finite λ because $S'(\lambda) = (n-d)^{-1} \sum_{\delta_j > 0} \delta_j / (1 + \lambda\delta_j)^2 > 0$. The normalization changes the scale of the original index but not the ordering of smoothness levels Guerrero (2008).

3 Classical parameter selection and numerical implementation

3.1 Selection criteria

For a fixed numerical value of μ, define $e_\lambda = y - \widehat{\tau}_{\lambda,\mu}$, $\operatorname{RSS}_\lambda = \|e_\lambda\|_2^2$, $h_{ii} = (H_\lambda)_{ii}$, and $\nu_\lambda = \operatorname{tr}(H_\lambda)$. The published classical selectors studied in Cortés-Toto et al. (2017)

are represented, up to constants irrelevant to an optimizer, by

$$\mathrm{CV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_{\lambda,i}}{1-h_{ii}}\right)^2, \tag{8}$$

$$\mathrm{GCV}(\lambda) = \frac{\mathrm{RSS}_\lambda/n}{(1-\nu_\lambda/n)^2}, \tag{9}$$

$$\mathrm{AICc}(\lambda) = \log(\mathrm{RSS}_\lambda/n) + \frac{2\nu_\lambda+1}{n-\nu_\lambda-2}, \tag{10}$$

$$\mathrm{BIC}(\lambda) = \log(\mathrm{RSS}_\lambda/n) + \frac{\nu_\lambda \log n}{n}. \tag{11}$$

The software checks denominators and excludes undefined candidates, evaluates a declared positive grid of penalty values, and compares an exact infinite-penalty candidate. In particular, a near-interpolating BIC minimum can be driven by the log(RSS) boundary, as our example illustrates. We do not interpret a finite-grid BIC minimum close to zero as a universally reliable statistical choice.

Guerrero Guerrero (2007) also considers the feasible plug-in $\widehat{\mu} = \mathrm{mean}(D_d y)$. This option is available *for fitting*; however, the formula for leave-one-out CV in (8) presumes that μ remains fixed under deletion. The current public selectors intentionally require a fixed numerical μ. Likewise, $\mathrm{tr}(H_\lambda)$ is a conditional EDF and is not automatically the full degrees of freedom of a procedure that estimates μ from the same observations.

3.2 Smoothing correction, inversion, and exact endpoints

With the full orthogonal eigenbasis of the penalty, $Q_d = V\,\mathrm{diag}(\delta_1,\ldots,\delta_n)V^\top$, the $n-d$ positive eigenvalues are $\delta_j = \sigma_j(D_d)^2$, and the remaining d are zero. Consequently, the spectral eigenvalues of $H_\lambda = (I+\lambda Q_d)^{-1}$ are

$$\alpha_j(\lambda) = \begin{cases} (1+\lambda\delta_j)^{-1}, & 1 \le j \le n-d, \\ 1, & n-d < j \le n. \end{cases} \tag{12}$$

This exposes both the exact limiting polynomial directions and the scalar attenuation of the penalized directions. If $\nu(\lambda) = \mathrm{tr}(H_\lambda) = d + \sum_{j=1}^{n-d}\alpha_j(\lambda)$, the original trace-based Guerrero index

$$S_G(\lambda) = 1 - \frac{\nu(\lambda)}{n} \quad \text{ranges from } 0 \text{ to } 1-\frac{d}{n}. \tag{13}$$

The upper bound is $1-d/n$, *not* $1-n/d$. The library also uses a normalized correction:

$$S(\lambda) = \frac{S_G(\lambda)}{1-d/n} = \frac{n-\nu(\lambda)}{n-d} \in [0,1]. \tag{14}$$

The map is strictly increasing for $\lambda \in [0, \infty)$ because

$$S'(\lambda) = \frac{1}{n-d} \sum_{j=1}^{n-d} \frac{\delta_j}{(1+\lambda\delta_j)^2} > 0. \tag{15}$$

The exact endpoints are $S(0) = 0$ and $S(\infty) = 1$; for any interior target, a one-dimensional bracketed solver inverts this map. Equal normalized smoothness makes trace scales comparable across d; it does not make the underlying trend geometries or roughness penalties identical.

3.3 Where the computational acceleration comes from

For fixed μ put $p_t = \mu\binom{t}{d}$ and $c = V^\top(y - p)$. One common projection computes c; after that, for each candidate λ,

$$\operatorname{tr}(H_\lambda) = d + \sum_{j=1}^{n-d} \alpha_j(\lambda), \tag{16}$$

$$\|y - \widehat{\tau}_{\lambda,\mu}\|^2 = \sum_{j=1}^{n-d} (1 - \alpha_j(\lambda))^2 c_j^2. \tag{17}$$

Thus a sweep of M values can obtain trace, normalized smoothness, GCV, AICc, BIC and related scalar profiles in $O(Mn)$ *after* the shared eigendecomposition and $O(n^2)$ projection. A full reconstructed trend remains $O(n^2)$ per penalty, and exact ordinary CV leverage diagnostics also involve the spectral basis. The full dense SVD has $O(n^3)$ preprocessing time and $O(n^2)$ storage. This decomposition/reuse trade-off explains the potential responsiveness of repeated-criteria explorations without claiming unconditional superiority to banded linear solvers Weinert (2007); Biessy (2026).

Large batches of spectral attenuation operations are naturally vectorizable. Dense linear algebra can also benefit from optimized multithreaded BLAS on CPUs. In principle, a GPU backend could parallelize matrix decompositions and batched matrix operations, but host/device transfer, synchronization, and initialization costs may negate a benefit for short series. We found `numpy` and `scipy.linalg.svd` in the package currently reviewed, not a released GPU solver; therefore no GPU speedup, power saving, or measured device execution is reported. A future benchmark should record CPU and GPU models, precision, BLAS and accelerator versions, transfer time, warmups, memory, and the number of candidate penalties.

3.4 Reusable computation and cost

The package exposes a reusable `TrendSmoother(n, order=d)` object. Construction forms the difference operator and its SVD. For each candidate λ, a fitted curve is obtained with the transformed vector c and spectral attenuation in (6). The library offers `fit`, `degrees_of_freedom`, `smoothness`, and `select_lambda`, while plotting

utilities visualize fits and criterion profiles. Sphinx incorporates executable Jupyter notebooks, with deterministic random seeds and figure code for inspection.

The computational trade-off must distinguish decomposition from reuse. A dense full SVD costs $O(n^3)$ preprocessing and $O(n^2)$ memory; after projection, EDF, smoothness, GCV, AICc, and BIC summaries cost $O(n)$ per penalty candidate, while full reconstruction and exact CV leverage scores require $O(n^2)$ each. In contrast, the banded structure of $I_n + \lambda Q_d$ allows efficient specialized direct solvers Weinert (2007). Thus the phrase *fast computation* means repeated spectral evaluation under a stated workload, not a blanket claim that SVD beats banded factorization. A separate reproducible benchmark in the repository measures setup and fit times independently.

4 Gaussian marginal likelihood and profiling the difference mean

4.1 Likelihood selection for fixed μ

A probabilistic interpretation of the penalty assumes observation noise $\varepsilon \sim N(0, \sigma^2 I_n)$ and stochastic differences $D_d\tau = \mu\mathbf{1}+\eta$ with $\eta \sim N(0, \sigma^2\lambda^{-1}I_{n-d})$. The d unpenalized polynomial coordinates act as nuisance fixed effects. In the $r = n - d$ positive-penalty singular coordinates $z_j = v_j^\top y$, write $p_t^{(1)} = \binom{t}{d}$, $b_j = v_j^\top p^{(1)}$. Marginalization gives

$$z_j \mid \mu, \lambda, \sigma^2 \sim N\left(b_j\mu, \sigma^2 c_j(\lambda)\right), \quad c_j(\lambda) = 1 + \frac{1}{\lambda\delta_j}. \tag{18}$$

This is a Gaussian residual/marginal-likelihood interpretation, not a new form of PLS. It is related to likelihood-based Whittaker–Henderson smoothing Biessy (2026). Profiling the variance σ^2 for a fixed numerical μ leaves, up to an additive constant,

$$J(\lambda; \mu) = r \log\left(\frac{1}{r}\sum_{j=1}^{r}\frac{(z_j - b_j\mu)^2}{c_j(\lambda)}\right) + \sum_{j=1}^{r}\log c_j(\lambda). \tag{19}$$

An independently developed branch of the package, `feature/mle-smoothness-and-data-mu-20261021`, adds this profiled Gaussian score for fixed μ or the conditional plug-in $\mu = \overline{D_d y}$. That plug-in must not be mislabeled as a joint MLE.

4.2 A closed-form estimator of μ conditional on λ

Let $w_j(\lambda) = c_j(\lambda)^{-1} = \lambda\delta_j/(1 + \lambda\delta_j)$. For a fixed positive λ, the generalized least-squares optimum satisfies

$$\widehat{\mu}(\lambda) = \frac{\sum_{j=1}^{r} w_j b_j z_j}{\sum_{j=1}^{r} w_j b_j^2}. \tag{20}$$

The denominator is positive because $D_d p^{(1)} = \mathbf{1}$. Equation (20) is also obtained by minimizing the PLS objective jointly in τ and μ for a fixed λ. Substitution in (19)

Table 1 Selected penalty and normalized smoothness for the case in Fig. 1. Trend RMSE is computed against the known latent simulation only.

Criterion	$\widehat{\lambda}$	S	Trend RMSE
CV	6.17	0.777	0.242
GCV	6.14	0.777	0.242
AICC	9.74	0.805	0.248
BIC	2.45e-07	0.000	0.454

gives the one-dimensional concentrated score $J(\lambda; \widehat{\mu}(\lambda))$, whose optimization produces a joint profile-likelihood candidate $(\widehat{\lambda}, \widehat{\mu})$. Such a solution depends on the Gaussian variance model, may lie near a boundary, and must be checked for identifiability and other local minima. This paper provides the equation and a separately reproducible numerical test; joint μ profiling is not yet an exposed release API.

In fact, taking the limit $\lambda \downarrow 0$ using $w_j \sim \lambda\delta_j$ proves

$$\lim_{\lambda\downarrow 0} \widehat{\mu}(\lambda) = \frac{(p^{(1)})^\top Q_d y}{(p^{(1)})^\top Q_d p^{(1)}} = \frac{\mathbf{1}^\top D_d y}{n-d} = \overline{D_d y}. \tag{21}$$

For positive λ the two expressions differ in general. Estimating μ adds a rank-one adjustment to the data-to-trend linear operator, so the fixed-μ effective degrees of freedom and ordinary leave-one-out denominators are not automatically valid after joint estimation. Prior literature already considers nonzero difference means Guerrero (2007); Cortés-Toto et al. (2017); the implementation contribution here is an explicit, testable profiling workflow, not a claim to have first introduced μ or likelihood selection.

5 Reproducible numerical illustration

5.1 Source-inspired nonlinear design

To demonstrate the interface without introducing an external data source, we reproduce the nonlinear simulation family of Cortés-Toto et al. Cortés-Toto et al. (2017):

$$\tau(t) = 0.6 f_{\text{Beta}(30,17)}(t/n) + 0.4 f_{\text{Beta}(3,11)}(t/n), \quad t = 1, \ldots, n. \tag{22}$$

We take $n = 50$, add independent normal errors with standard deviation 0.5, omit seasonality, and fix the pseudo-random seed at 19. These choices define one illustrative replication, not a full reproduction of the article's factorial simulation. We fit $d = 2$, $\mu = 0$, and select λ from the same 121-point geometric grid augmented with the exact $\lambda = \infty$ candidate under each criterion. The latent trend in (22) is known, allowing an *ex post* trend-recovery RMSE; it is not a future forecast error.

As shown in Table 1, CV and GCV agree in this particular realization, whereas AICc selects a somewhat smoother fit. BIC selects a penalty extremely close to zero on the inspected finite grid; its apparent interpolation tendency illustrates a known

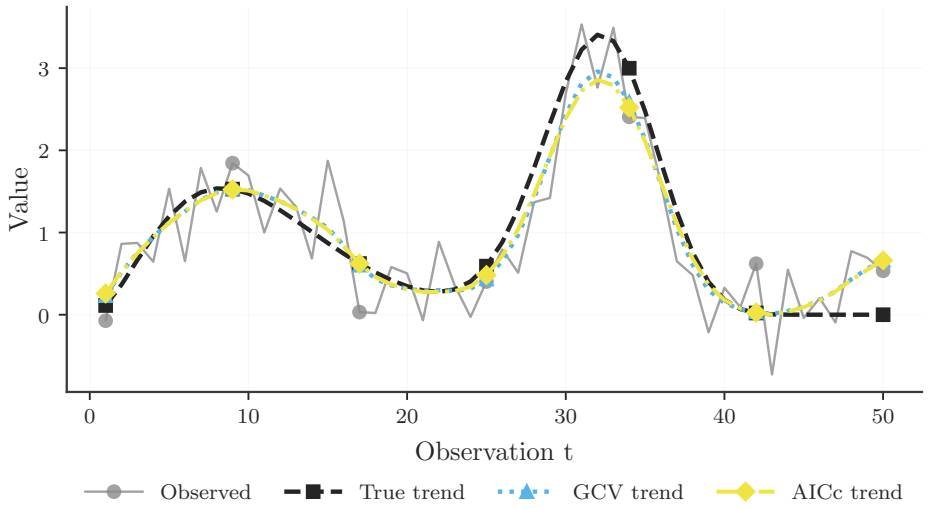


Fig. 1 A synthetic nonlinear case based on Cortés-Toto et al. Cortés-Toto et al. (2017). The curves show *in-sample* estimates and known simulated truth, not out-of-sample forecasts. Source data are regenerated by `figures/make_figures.py`.

practical boundary concern and is not evidence that BIC is generally ineffective. The simulated recoveries are descriptive; no general method ranking follows from a single seed.

5.2 Algebraic and numerical checks

An independent check compares the SVD implementation with the direct normal equations for $n = 50$, $\lambda = 30$, $d \in \{1, 2, 3\}$ and $\mu \in \{0, 0.15\}$ on a second fixed-seed Gaussian sample. The maximum componentwise absolute difference is 2.01×10^{-12} across these six cases, and the limiting constraint $D_d \widehat{\tau}_{\infty,\mu} = \mu \mathbf{1}$ holds to maximum error 4.42×10^{-12}. Both values result from the supplied deterministic regression script in IEEE double precision; they verify the tested algebra rather than establish stability for all n, d, λ. Higher differencing orders can be ill-conditioned, and systematic machine-dependent speed measurements remain a distinct stage of the study.

6 Measured computational cost and repeated parameter evaluation

6.1 Experimental design and independent baselines

Numerical correctness and elapsed time are distinct properties. We therefore benchmark the same PLS estimator at matched (n, d, μ) and identical candidate penalties, using four implementations: (i) a deliberately naive dense solver that reconstructs D_d,

Table 2 Leading arithmetic complexity for M reconstructed trends of length n and fixed order d. Setup is counted and no exact operation counts or GPU measurements are asserted.

Method	Setup	Repeated full fits
Fresh dense solves	rebuilt per λ	$O(Mn^3)$
Cached dense Q_d	dense penalty matrix	$O(Mn^3)$
Spectral SVD	$O(n^3)$	$O(Mn^2)$
Direct banded solver	$O(nd^2)$	$O(Mnd^2)$

Table 3 Host-measured end-to-end median times in milliseconds, including one-time setup, for M full fitted trends. Every method computes the same trends for the same n, $d = 2$ and $\mu = 0$; R is the naive-fresh-dense time divided by the spectral time. Values and the numerical parity check are generated from the raw CSV files, not entered manually.

n	M	Fresh	Cached	Spectral	Banded	R
40	1	0.029	0.027	0.204	0.123	0.14
40	64	1.388	0.739	1.302	1.032	1.07
100	1	0.075	0.073	0.801	0.281	0.09
100	64	4.589	2.642	2.013	1.532	2.28
200	1	0.471	0.350	3.580	0.535	0.13
200	64	20.984	10.348	4.982	2.017	4.21
400	1	3.192	3.217	20.165	1.103	0.16
400	64	196.554	106.894	20.885	3.188	9.41

$Q_d = D_d^\top D_d$, and solves the normal equations at every candidate; (ii) a denser but fairer baseline that caches Q_d and repeats its dense solve; (iii) the reusable full-SVD spectral implementation of `TrendSmoother`; and (iv) a banded solver that assembles Q_d directly in its sparse-band storage and uses a structure-aware linear solver. The last comparison is crucial: beating dense normal equations does not imply beating specialized PLS solvers Weinert (2007).

For every case a deterministic Gaussian sequence is generated, μ is held at either zero or a fixed nonzero value, and λ spans 10^{-3} to 10^3 on a geometric grid. The `quick` experiment varies $n \in \{40, 100, 180\}$, $M \in \{1, 16, 64\}$ with $d = 2$ and adds $d \in \{1, 3, 5\}$ at $n = 100$, $M = 24$, together with a nonzero-μ case. The `full` experiment reaches $n = 400$, $M = 256$ and additional difference orders. All results include one-time setup (not only post-SVD solves); we record medians, extrema, and median absolute deviations over repeated independent runs. We bound maximum disagreement against an independent banded solve for *every* tested penalty. Runs use a declared random seed, source revision, machine metadata, NumPy/SciPy versions and a stated BLAS-thread limit. The code and machine-readable outputs are confined to `paper/benchmarks/`.

Table 4 Median end-to-end times (milliseconds) to evaluate M effective-degrees-of-freedom traces, including setup. These entries measure only scalar $\mathrm{tr}(H_\lambda)$ quantities, not reconstructed trends.

n	M	Fresh	Cached	Spectral	Banded
40	8	0.213	0.150	0.203	0.322
100	8	1.059	0.832	0.801	1.072
100	32	4.220	3.307	0.885	3.615
180	32	15.914	12.208	2.830	10.114
250	64	92.614	59.421	6.090	38.391

6.2 End-to-end full-fit timing results

The crossover between methods should be evaluated using *end-to-end* times, including the SVD setup cost, for both a single smoothing parameter and a repeated-λ sweep. A ratio greater than one denotes a speed advantage over the naive baseline on the reported machine, not a universal speedup. In particular, the structured banded baseline can outperform a reusable SVD for full trend reconstruction.

6.3 Scalar smoothness and EDF are different workloads

To evaluate $\mathrm{tr}(H_\lambda)$ repeatedly, the spectral method needs only $n - d$ positive eigenvalues:

$$\mathrm{tr}(H_\lambda) = d + \sum_{j=1}^{n-d} (1 + \lambda\delta_j)^{-1}.$$

This is $O(Mn)$ work *after* spectral setup, not $O(Mn^2)$ reconstructed trends. An independent second experiment compares repeated dense computations of $\mathrm{tr}[(I + \lambda Q_d)^{-1}]$, a banded solver applied to all n canonical right-hand sides, and the spectral sum. The experiment checks algebraic equivalence separately from timed regions. Even when the spectral trace is the most efficient *marginal* score after setup, the $O(n^3)$ SVD cost can still dominate small workloads.

Measurement provenance: Platform: Windows-10-10.0.26100-SP0; CPU: Intel64 Family 6 Model 183 Stepping 1, GenuineIntel; Logical CPUs: 32; NumPy: 2.4.6; SciPy: 1.17.1; BLAS threads: 1; Repetitions: 7; Preset: full; Revision: local-unrecorded.
Maximum observed absolute trend disagreement: $4.32e-10$; maximum EDF trace disagreement: $5.48e-12$.

6.4 Interpreting speed and computational cost

A timed speed ratio is defined by the two end-to-end medians, for example $R_{\mathrm{dense/spectral}} = t_{\mathrm{dense}}/t_{\mathrm{spectral}}$. A value above one favors the spectral approach for the tested workload, not for arbitrary data lengths or hardware. CPU time, wall time, arithmetic complexity and energy consumption are different quantities. CPU numerical libraries already use optimized native linear algebra, but this experiment does

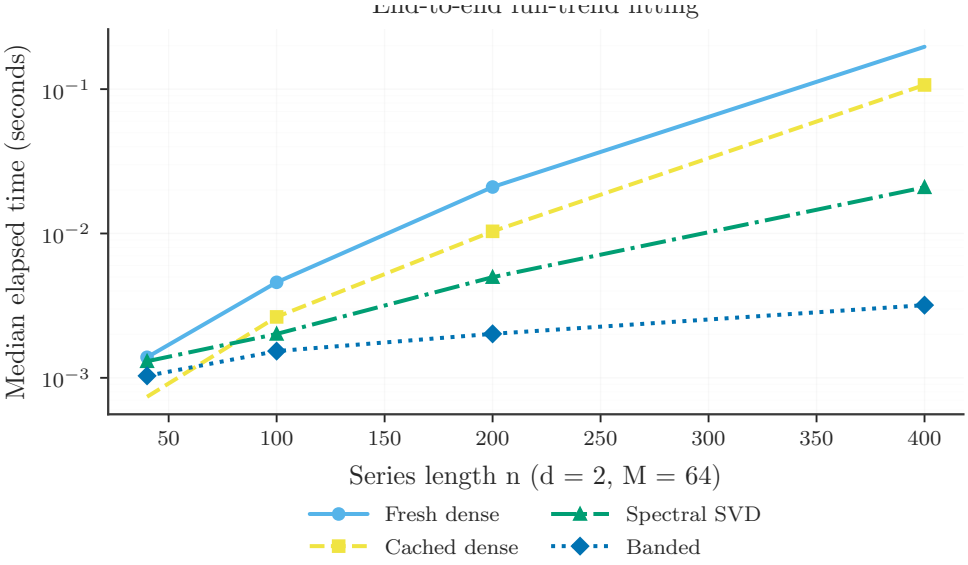


Fig. 2 End-to-end median times for $M = 64$ fitted curves, including setup, generated on the executing host from the raw CSV files. These are hardware-specific measurements, not an assumption that spectral reconstruction is the universally fastest method.

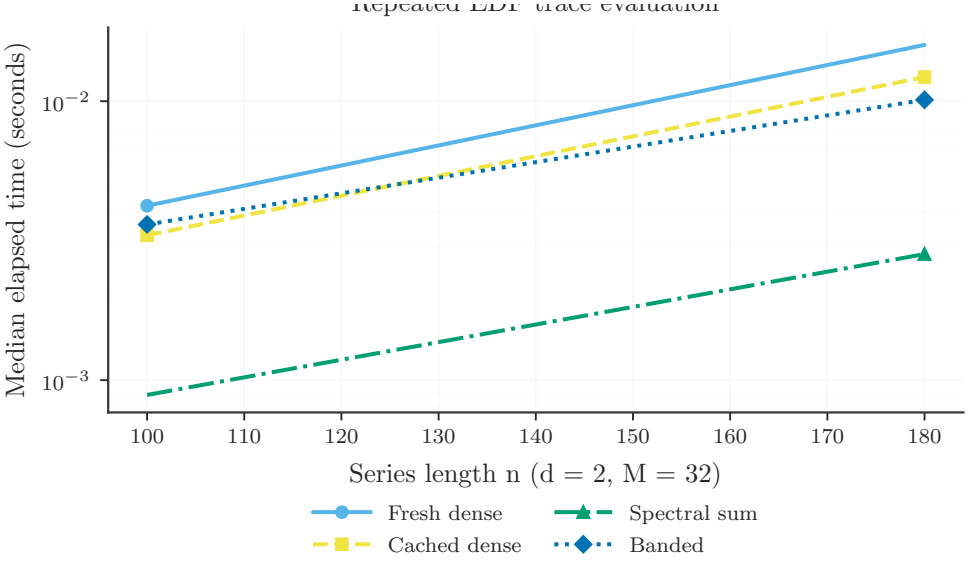


Fig. 3 Repeated effective-degrees-of-freedom trace evaluations ($M = 32$), including preprocessing. The quantity is scalar $\mathrm{tr}(H_\lambda)$, not a full trend reconstruction.

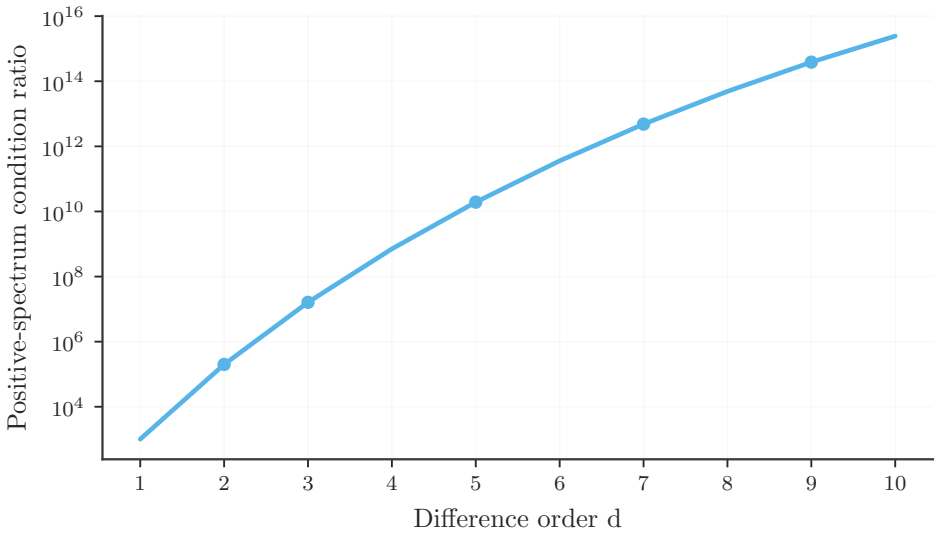


Fig. 4 Condition ratio of the positive spectrum of $D_d^\top D_d$ for $n = 50$, $d = 1, \ldots, 10$. Values are recomputed from singular values of D_d; they do not represent hardware-specific runtime.

not establish GPU execution or electricity use. An explicit device and power measurement would be necessary to make either claim. Results are intentionally generated only through a manual command, so the manuscript does not inadvertently contain timings from a different machine presented as those of its authors.

7 A controlled exploration of larger difference orders

The software accepts any integer $0 \leq d < n$, but this algebraic admissibility must be distinguished from floating-point stability. We therefore conduct a deterministic diagnostic with $n = 50$, seed 24021, difference orders $d = 1, \ldots, 10$, $\mu = 0$, and common normalized smoothness $S = 0.86$. At each order, we form the full SVD of D_d, compute the condition ratio

$$\kappa_+(Q_d) = \frac{\max_{1 \leq j \leq n-d} \delta_j}{\min_{1 \leq j \leq n-d} \delta_j}, \tag{23}$$

and compare the spectral fitted trend with an independent direct solution of $(I + \lambda Q_d)\tau = y$. The ratio is calculated from the *strictly positive* penalty eigenvalues; the full Q_d has zero eigenvalues and is singular for every $d \geq 1$.

The ratio rises from about 10^3 at $d = 1$ to 2.44×10^{15} at $d = 10$. At $d = 10$, the maximum componentwise difference between the SVD-based fit and the direct normal-equation solution is approximately 3.36×10^{-7} for the declared sample and smoothing target, versus approximately 1.14×10^{-15} at $d = 1$. The direct normal-equation result is not presumed an exact ground truth at large condition numbers: both approaches are affected by finite precision. The experiment illustrates why support

for many difference orders needs accompanying conditioning diagnostics, rather than establishing new high-order theory, unconditional stability, or performance superiority. Machine-specific setup/fitting times are separately exported by the experiment script for later audit; they are not promoted to general speed benchmarks.

8 Contemporary empirical dashboards

8.1 Sources, competing criteria and training-origin design

The real-data illustration uses the frozen NOAA Global Monitoring Laboratory Mauna Loa monthly-mean atmospheric CO_2 concentrations from January 1990 through September 2026 ($n = 441$, last reported value 426.07 ppm) NOAA Global Monitoring Laboratory and Scripps Institution of Oceanography (2026), and the monthly US effective federal funds rate FEDFUNDS from January 2010 through September 2026 ($n = 201$, last reported value 3.75 percent) Board of Governors of the Federal Reserve System (2026). The FEDFUNDS observations are monthly averages of daily effective rates, not seasonally adjusted. We also repeat the CO_2 analysis on its last 120 observed months to examine sensitivity to the estimation window. All raw data and publisher vintages are frozen in the accompanying source repository; neither dataset contains invented October 2026 observations.

At each difference order $d \in \{1, 2, 3, 4\}$ we evaluate six procedures: classical leave-one-out CV, GCV, AICc, BIC, a Gaussian marginal profile likelihood criterion (MLE), and user-controlled normalized smoothness $S = 0.86$. The first five criteria each minimize their own in-sample score over the *same* 29 candidate penalties obtained by inverting the grid

$$\mathcal{S} = \{0.10 + 0.88j/28 : j = 0, \ldots, 28\}, \quad \lambda_j = S^{-1}(s_j).$$

Thus, differences among those selections at a fixed d and μ do not arise from different candidate penalty grids. The bounded grid cannot certify a global optimum, and an optimum on its lowest or highest candidate is a *grid boundary* result. Controlled $S = 0.86$ is a prescribed smoothness level, *not* a sixth optimization criterion.

8.2 Reference difference mean and conditional likelihood

We repeat the analysis under two specifications:

$$\mu_0 = 0, \qquad \widehat{\mu}_{d,\mathcal{T}} = \frac{1}{|\mathcal{T}| - d} \mathbf{1}^\top D_d y_{\mathcal{T}},$$

where $\mathcal{T}$ is the *available training window* at the relevant origin. The second value depends on the difference order and is not a universal drift parameter shared across orders. For CV, GCV, AICc and BIC, we treat the numeric $\widehat{\mu}_{d,\mathcal{T}}$ as *fixed conditional on that training sample* when scoring candidate penalties. In particular the reported CV is not the exact leave-one-out score for a procedure that re-estimates μ in every deletion. The MLE panel uses the current library's `mu="data"` plug-in: its Gaussian marginal likelihood is profiled over variance conditional on this reference mean, *not*

jointly maximized over (λ, μ). For $\mu = 0$ the same MLE criterion is evaluated with that fixed value. All results report both the numeric μ and the selected pair $(\widehat{\lambda}, S(\widehat{\lambda}))$.

8.3 Historical 12-month holdout dashboard

We reserve the final 12 complete observed months of each series. For every row/column combination, its reference mean, penalty and PLS trend are fitted using the earlier training segment *only*. Rows correspond to $d = 1, 2, 3, 4$ and columns to CV, GCV, AICc, BIC, MLE, and controlled smoothness. Panel-level RMSE and MAE compare the resulting 12-month *trend extrapolations* with the held-out raw observations. The selected penalties and the numeric plug-in means are recomputed when the origin changes.

8.4 Full-sample future trend extrapolation

With the final data month included, we fit each smoother again and extend the estimated trend for 12 future calendar months (October 2026–September 2027). The PLS estimator is two-sided on observed data; it does *not* intrinsically define future observations. We therefore use an explicit tail rule: regress the last up to 36 fitted monthly trend values on a locally centered polynomial of degree $\min(d-1, 2)$, shift that polynomial to pass through the final fitted trend, and extrapolate it one to 12 steps ahead. $d = 1$ thereby produces a constant continuation, $d = 2$ a locally linear one, and $d \geq 3$ a quadratic continuation under the stated cap. No future seasonality, innovations, or prediction intervals are modelled.

8.5 Smoothness, likelihood and holdout error

Figure 7 reports the selected raw penalty, its normalized smoothness, and historical holdout RMSE and MAE across orders. The penalty λ and the normalized trace coordinate $S(\lambda)$ describe *the same fit*, not competing estimates. Unlike the penalty, a common target S normalizes the fraction of penalizable degrees of freedom across d; it does not make the geometric trend shapes or extrapolation rules equal. The MLE column differs from CV/GCV/AICc/BIC through its probabilistic profile-likelihood criterion; its numerical score is not directly comparable with their objective values.

8.6 Sensitivity to data-derived nonzero means

The alternative reference mean must be estimated separately for each $(d, \mathcal{T})$ and thus can alter fitted endpoints, selected smoothness, and subsequent extrapolation. We compare the two specifications without reusing a mean estimated on the holdout observations.

8.7 Economic-series comparison and supplementary figures

The same complete factorial set of 24-panel dashboards is reproduced for the US federal funds rate, and for the last 120 months of Mauna Loa CO_2, with both μ

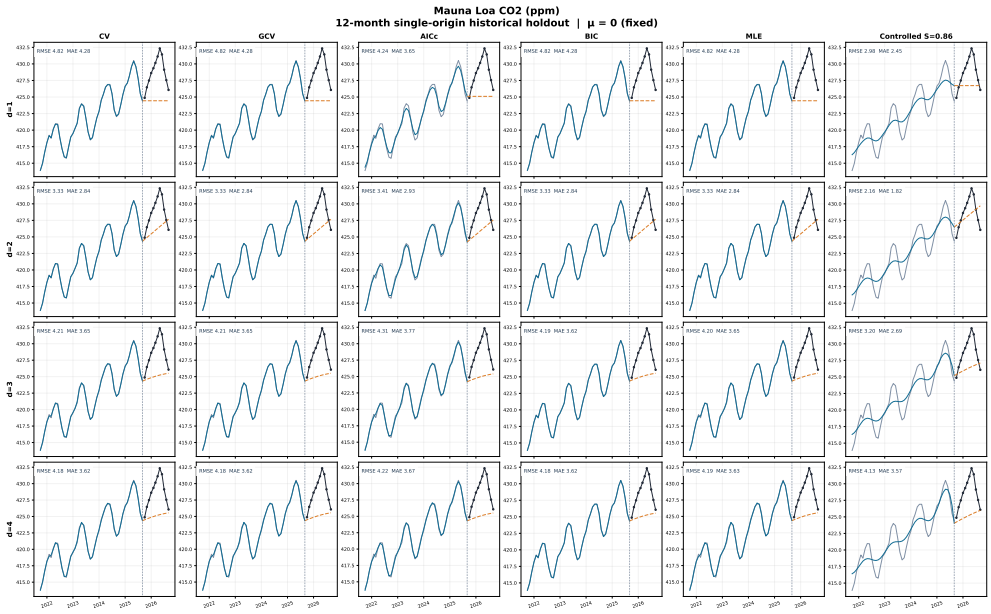


Fig. 5 Mauna Loa CO_2 historical 12-month single-origin holdout with fixed $\mu = 0$. Rows are difference orders $d = 1, 2, 3, 4$ and columns are CV, GCV, AICc, BIC, MLE, and prescribed $S = 0.86$. Gray curves represent observed training data, blue curves are fitted historical PLS trends, black markers are the held-out observations, and dashed orange curves are 12-step extrapolations constructed without knowledge of that holdout. Full-resolution vector panels accompany the source code. Scores are in-sample criteria; holdout losses measure the extra extrapolation procedure, not a separately specified stochastic forecast model.

Table 5 Mauna Loa CO_2: all six selection procedures for $d = 2$, showing fixed-zero versus data-derived reference differences. These are one-origin descriptive holdout losses (ppm). AICc, BIC, CV, GCV and MLE select from the same finite penalty grid; Controlled is prescribed.

μ mode	Selector	μ	λ	S	RMSE	MAE
0 fixed	CV	0	0.0205	0.100	3.328	2.839
0 fixed	GCV	0	0.0205	0.100	3.328	2.839
0 fixed	AICc	0	0.428	0.509	3.413	2.930
0 fixed	BIC	0	0.0205	0.100	3.328	2.839
0 fixed	MLE	0	0.0205	0.100	3.328	2.839
0 fixed	Controlled S=0.86	0	41.8	0.860	2.162	1.820
training plug-in	CV	-0.0062	0.0205	0.100	3.328	2.839
training plug-in	GCV	-0.0062	0.0205	0.100	3.328	2.839
training plug-in	AICc	-0.0062	0.428	0.509	3.414	2.931
training plug-in	BIC	-0.0062	0.0205	0.100	3.328	2.839
training plug-in	MLE	-0.0062	0.0205	0.100	3.328	2.839
training plug-in	Controlled S=0.86	-0.0062	41.8	0.860	2.168	1.825

specifications. Figure 9 displays the economic-series holdout for $\mu = 0$; the remaining full-resolution PDF/SVG/PNG panels and machine-readable per-cell results accompany the paper.

A single held-out block, especially for CO_2 with a pronounced annual cycle, cannot establish forecast superiority; selection procedures target different in-sample quantities and the extrapolation lacks a seasonal model. For inferential forecasting claims, further work requires multiple chronological origins, horizon-specific evaluation, paired uncertainty analysis, and explicit modelling of seasonal or policy-driven changes. No optimization strategy here is presented as universally superior solely because of this illustration.

9 Conclusions

We have presented an auditable implementation of established penalized least-squares smoothing, together with additional derivations of the difference nullspace, the corrected smoothness index, and a Gaussian mean-difference profile estimator. The latter has numerical checks but is not yet a released joint-MLE API. Our contemporary NOAA CO_2 and US federal funds illustrations now compare five in-sample optimization criteria (CV, GCV, AICc, BIC, and conditional Gaussian profile MLE) and a prescribed smoothness reference over four differencing orders, using both $\mu = 0$ and a training-sample-derived reference mean. The 24-panel dashboards expose the consequences for a single chronological holdout and subsequent trend extrapolation; paired (λ, S) selections, uncertainty about model selection and reference-mean handling are reported explicitly. A single historical holdout is not offered as a validated forecasting ranking. Spectral decomposition makes repeated penalty evaluation transparent and inexpensive after preprocessing; specialized banded solvers remain important competitors for isolated fits and large samples. The source-inspired experiment shows how the same observations can lead to distinct smoothness choices under established criteria, while deterministic checks support correctness in a bounded set of cases. The separate numerical comparison places naive dense normal equations, cached dense solves, reusable SVD, and an independently assembled banded solver under matched workloads. This distinction is important: spectral reuse is particularly attractive for repeated trace-based scores, but is not automatically optimal for reconstructing full trends. Broader performance conclusions require reported machine characteristics, workload-specific runtimes, and stability diagnostics. Neither high-order differencing nor likelihood-based smoothing is claimed as a newly discovered concept. No GPU acceleration has been verified in the released CPU implementation.

Declarations

Funding. To be confirmed by the authors before submission.
Competing interests. To be confirmed by the authors before submission.
Ethics approval and consent. Not applicable to the presented numerical algorithms and historical aggregate time-series examples; authors to verify.
Data availability. The simulated data can be reproduced from the supplied scripts and seeds. Current NOAA monthly CO_2 observations and US effective federal funds

rate observations from Federal Reserve/FRED are provided as dated, frozen CSV snapshots with source URLs and SHA-256 checksums, along with executable figure-generation code. The archived `statsmodels` 1990–1999 example is retained in the repository but is no longer the submission example.

Code availability. The implementation and paper-specific simulation and benchmark scripts are maintained at https://github.com/heri-espino/trendestimation. A permanently archived release identifier should be added before submission.

Author contributions. To be confirmed and approved by both authors before submission.

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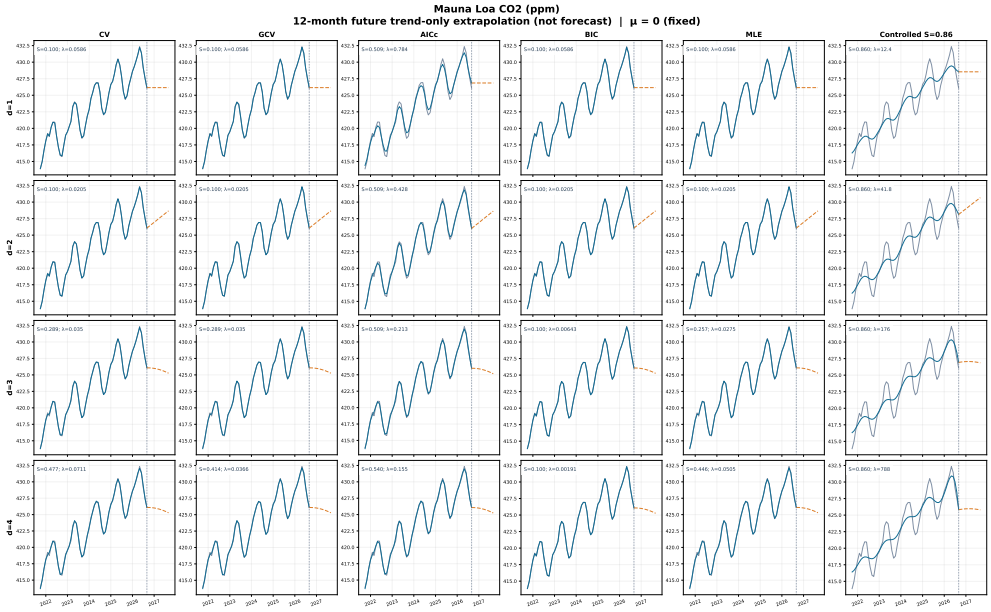


Fig. 6 Full-sample Mauna Loa CO_2 trend extensions under $\mu = 0$ in the same 4×6 method/order layout. Dashed segments lie *beyond* the September 2026 observed-data cutoff and are descriptive anchored polynomial extensions, not observed data, validated stochastic forecasts, or confidence bands.

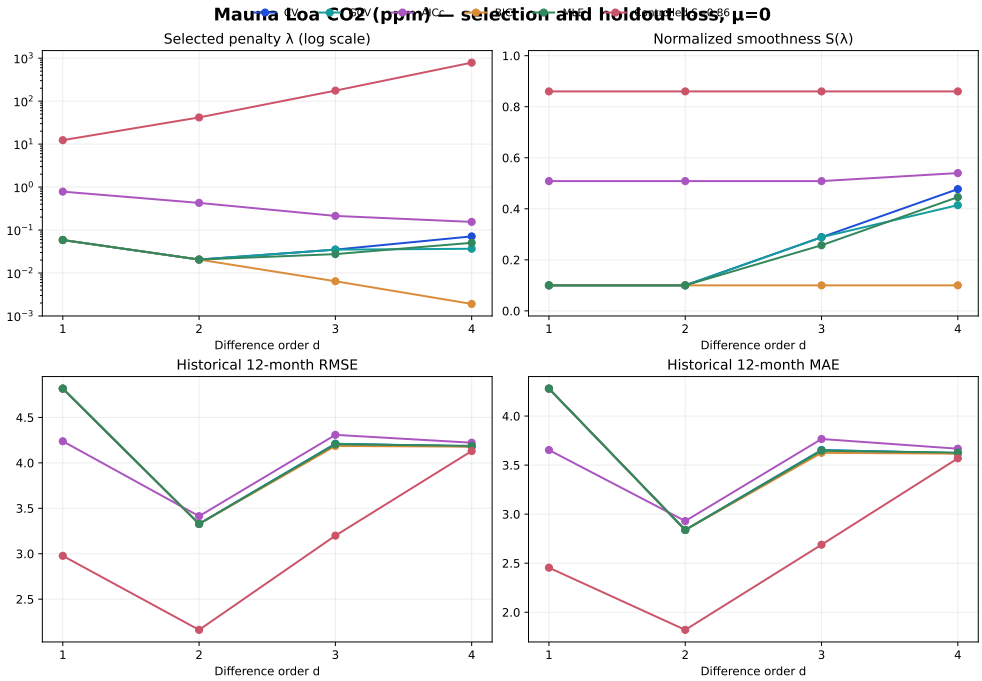


Fig. 7 Mauna Loa CO_2 with $\mu = 0$: selected λ (log scale), normalized S, and 12-month holdout RMSE and MAE by differencing order and selection method. These are descriptive single-origin losses, not a test of a calibrated future-data predictive distribution.

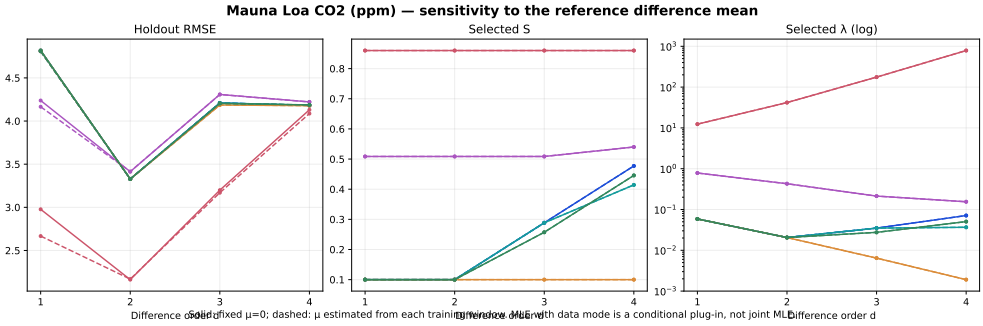


Fig. 8 Mauna Loa CO_2: comparison of the fixed $\mu = 0$ (solid curves) and data-derived μ (dashed) across d for all six procedures. Panels report historical holdout RMSE, selected normalized smoothness and selected penalty. The dashed results condition on the data-derived reference difference mean, including for MLE; they do not represent joint likelihood estimation of μ and λ.

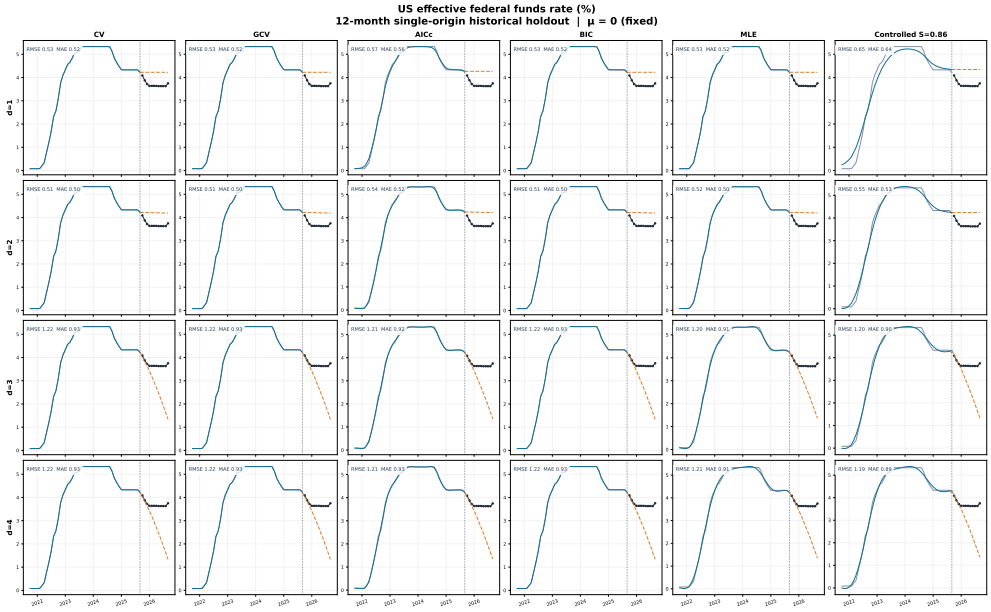


Fig. 9 Monthly US effective federal funds rate: 12-month historical holdout with $\mu = 0$; rows $d = 1, 2, 3, 4$, columns CV/GCV/AICc/BIC/MLE/ Controlled. The economic series is not seasonally adjusted. The dashed segments are the same anchored trend-extrapolation rule applied at the training origin.